Context Switching Lab in Python

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1 Objective

The objective of this lab is to implement a context switching mechanism in Python, simulating processes that execute different operations based on instructions from specified from list, tuples, list of tuples and or text files.

Introduction

This Lab Manual illustrates how to implement context switching in Python using threading, and further discusses methods to control the execution sequence of threads. Further, we can employ various synchronization mechanisms, such as Locks and Events, to manage thread execution order effectively.

2 Student's Homework

2.1 Without Threading

Create five input files which contain various arithmetic operations (instruction sets in different files) as specified in the following format:

- `file1.txt`

  ```
  ADD a b AddResult
  SUB c d SubResult
  ```

- `file2.txt`

  ```
  MUL e f MulResult
  DIV g h DivResult
  MOD i j ModResult
  ```

- `file3.txt`

```
DIV c d DivResult
MUL e f MulResult
SUB g h SubResult
```

- file4.txt

```
ADD i j SumResult
SUB b a SubResult
MUL c d MulResult
DIV e f DivResult
```

- file5.txt

```
ADD AddResult 1000 MulResult
```

2.2 With Threading

Repeat above now with threading.

Critical Thinking

After overall implementation, answer the following questions:

- Compare the performance of threaded vs non-threaded versions
- How would priority scheduling affect your implementation?

3 Submission

Prepare a document and outputs as screenshots and printout for submission. Submission due date is 16th April 8:00 AM in the OS theory class.